

Wilson-Cowan equations

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The Wilson–Cowan model describes the dynamics of interacting excitatory and inhibitory neuron populations. In this study, we explore the fundamental behavior of this model, beginning with a simplified single-region system. Through linear stability analysis and bifurcation theory, we identify distinct dynamical regimes (quiescent, oscillatory, and bistable) and classify their transitions via Hopf, Pitchfork, Saddle-Node, and Takens–Bogdanov bifurcations.

Building on this, we expand the model to a network of interconnected brain regions using structural data encoded in an adjacency matrix. By computing the Adjacency Matrix eigenvalues, we uncover universal patterns in healthy brains. By solving the eigenvalue problem for the corresponding connectome Laplacian, we the harmonic modes of brain organization (eigenvalues). We use these to analyze whole-brain dynamics, quantifying synchronization across regions using the Kuramoto order parameter.

Additionally, we consider stochastic extensions of the Wilson–Cowan equations via a master equation approach and derive their corresponding Fokker–Planck and Langevin formulations.

I. INTRODUCTION

The Wilson-Cowan model was first introduced in 1972 by Hugh R Wilson and Jack D Cowan [1], in order to explain the brain dynamics held at the mesoscopic level.

In this work, we wish to study this model. As a first approach, we may analyze a simple version of the differential equations that dictate the interactions between the agents of the system. In order to do this, we aim to do a linear stability analysis, in order to find different dynamical regimes that arise with the variation of the synaptic weights. Between these regimes, we will find different types of bifurcations.

Once this analysis is done, we move on to analyze a wider spectrum of the brain, first focusing on the structure and characterizing different regions and the connection between them, making use of the adjacency matrix, from which we compute the Laplacian spectrum to analyze the different structural modes adopted by the brain. We also analyze the degree distribution and adjacency eigenvalues distribution across individuals, aiming to identify signatures of universality that are present in healthy brains.

As a final step, to understand global synchronization, we compute the Kuramoto order parameter, allowing us to study the emergence of the different dynamical regimes that manifest in the system.

II. WILSON-COWAN MODEL

In this model (see Figure 1), neural activity is represented by two populations of neurons, namely:

- **Excitatory:** These neurons excite others and themselves, resulting in an increased neural activity.
- **Inhibitory:** These neurons inhibit others and themselves, resulting in a decreased neural activity.

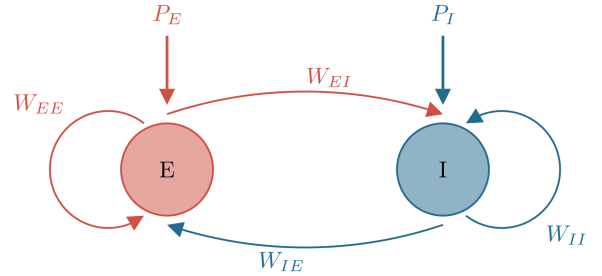


FIG. 1: Illustration of the Wilson-Cowan model.

For each population, there exists a dynamical non-linear differential equation. These equations can be either obtained phenomenologically, assuming mutual, self and external interactions, such as in the original work by Wilson and Cowan [1] or from a microscopic point of view, which involves a master equation that models stochastic interactions between single neurons, which retrieves the same phenomenological expressions via the Kramers-Moyal expansion [2–5], which is done in section B. One obtains, in any case:

$$\tau_E \frac{dE}{dt} = -E + S(w_{EE}E - w_{EI}I + P_E), \quad (1)$$

$$\tau_I \frac{dI}{dt} = -I + S_I(w_{IE}E - w_{II}I + P_I), \quad (2)$$

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where:

- $(E(t), I(t)) \equiv (E, I)$ are the average number of neurons firing per unit time.
- τ_E, τ_I are time constants which represent how quickly the population decays in absence of stimuli.
- w_{XY} are synaptic weights from population Y to X .
- $P_{E,I}(t) \equiv P_{E,I}$ are external inputs.
- $S(x)$ is a sigmoid function, which main goal is to model the non-linearity of the interactions and to saturate when the inputs surpass a certain threshold. The specific choice of the Sigmoid function varies the range of values E and I can take. In the usual literature, (see [1, 5]) the following function is used:

$$S(x) = \frac{1}{1 + e^{-\alpha(x-\theta)}} \quad (3)$$

Which gives values in the range $E, I \in [0, 1]$.

For this work, a different range is used, $E, I \in [-1, 1]$, however, this choice does not affect the dynamics of the system. The following Sigmoid function is used:

$$S(x) = \tanh(x). \quad (4)$$

A. Dynamical regimes and bifurcations

A simple version of the equations give the full scope of the dynamics of the system. In this simplification:

- We set $\tau_{E,I} = 1$.
- No external inputs are considered $P_{E,I} = 0$.

We wish to compute a phase diagram for the excitatory synaptic weights, namely, (w_{EE}, w_{IE}) for fixed values of the inhibitory, $(w_{II}, w_{EI}) = (1, 10)$, which were tuned according to [5].

In Figure 2, we find 3 distinct dynamical regimes:

- Figure 2.A, Quiescent: Stable fixed point where little to no activity or fluctuations is held over time.
- Figure 2.B, Oscillatory: Unstable fixed point; The system cycles indefinitely through dynamic states, limit cycle
- Figure 2.C, Bistable: Two stable fixed points. One unstable fixed point. The system settles in one of two states, but a perturbation can change the state due to the presence of the saddle node (see Figure 2).

Between the regimes, we find 4 distinct bifurcations, which analytical derivation can be found in section A.

- Hopf bifurcation: we find a constant function at the trivial fixed point $(E^*, I^*) = (0, 0)$, at $w_{EE} = 3$, the fixed point from the quiescent regime becomes unstable and oscillations begin.
- Pitchfork: we find a linear function at the trivial fixed point $(E^*, I^*) = (0, 0)$:

$$w_{IE} = \frac{w_{EE} - 1}{5}. \quad (5)$$

The stable fixed point from the quiescent becomes unstable and gives rise to two new stable symmetric fixed points (bistable).

- Saddle-Node: two fixed points (one stable, one unstable) collide and annihilate each other. It marks the limit between the oscillatory and the bistable regime.
- Takens-Bogdanov: point where a fixed point presents a double zero eigenvalue [6], indicating the simultaneous presence of a saddle-node and a Hopf bifurcation.

B. Structure of the brain: The adjacency matrix

Having studied the simplest model, which only covers the scope of a single region, we can move on to analyze the system for several regions, and the interactions between them.

For that purpose, we introduce the Adjacency matrix, a tool which contains the weights between different nodes of the brain, where each node represent a different region, hence showing correlation between the different parts of study. For simplicity, we will use binary weights, where 1 means connected and 0 unconnected. The resulting multi-region Wilson-Cowan equations are of the form:

$$\tau_E \frac{dE_i}{dt} = -E_i + S \left[\sum_{j=1}^N (A_{ij} w_{EE} E_j - w_{EI} I_j) + P_E \right], \quad (6)$$

$$\tau_I \frac{dI_i}{dt} = -I_i + S \left[\sum_{j=1}^N (A_{ij} w_{IE} E_j - w_{II} I_j) + P_I \right]. \quad (7)$$

Where we take into account the adjacency matrix only for the excitatory components, as we are working with varying excitatory synaptic weights and fixed inhibitory weights. Given the graph representation of the human connectome, we define the connectome Laplacian [7] as:

$$\Delta_G = \frac{1}{2} ((\mathbf{D} - \mathbf{A})^\top (\mathbf{D} - \mathbf{A})). \quad (8)$$

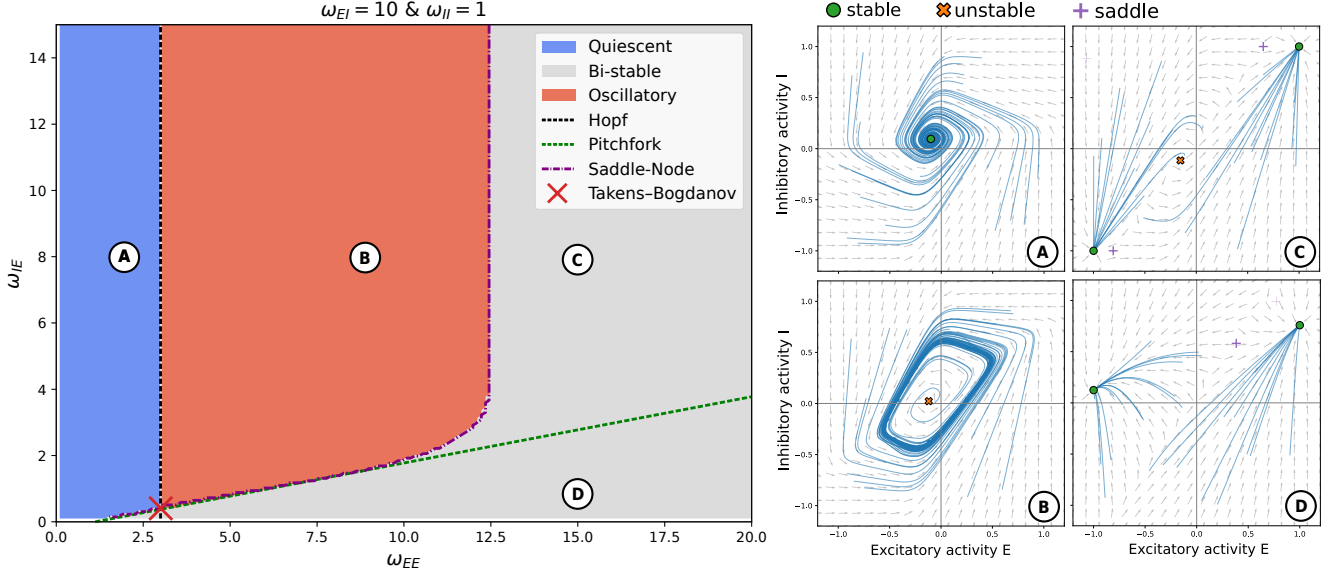


FIG. 2: **Dynamical regimes in the Wilson-Cowan model:** Left: Phase diagram of the Wilson-Cowan system in the $(\omega_{EE}, \omega_{IE})$ parameter space, with fixed $\omega_{EI} = 10$ and $\omega_{II} = 1$. Regions correspond to qualitatively different dynamical behaviors: Quiescent (no activity), Bi-stable (two stable fixed points), and Oscillatory (limit cycles). Curves indicate bifurcations: Hopf, Pitchfork, Saddle-Node, and Takens-Bogdanov. Right: Representative phase portraits illustrating the dynamics in each regime. (A) shows convergence to a single fixed point (Quiescent). (B) depicts stable oscillations (limit cycle). (C) and (D) displays coexisting stable fixed points (Bi-stable), with trajectories converging to different attractors depending on initial conditions. they differ in the number of saddle-nodes that redirects the system from one fixed point to another. Thus in (D) it is very unlikely to escape from the left-most stable point.

Here, $\mathbf{A}$ is the adjacency matrix and $\mathbf{D}$ is the degree matrix, defined elementwise by:

$$\mathbf{D}(i, i) = \sum_{j=1}^n \mathbf{A}(i, j), \quad 0 \text{ otherwise.} \quad (9)$$

The matrix $\Delta_{\mathcal{G}}$ is a symmetric and positive semi-definite operator that defines the local connectivity structure of the network. To extract the characteristic modes of the connectome, we solve the eigenvalue problem:

$$\Delta_{\mathcal{G}} \psi_i = \lambda_i \psi_i. \quad (10)$$

The resulting eigenvectors ψ_i are known as *connectome harmonics*. They form an orthogonal basis over the graph and play a role analogous to Fourier modes in signal processing [8]. The eigenvalues correspond to the allowed frequencies of standing waves that arise from the specific geometry, while the eigenvectors describe the spatial wave patterns or modes associated with those frequencies [9, 10].

In this way, connectome harmonics show the fundamental patterns of network organization over the whole scope of the brain. For a specific mode, the different regions of the brain will have a stronger or weaker contribution (see Figure 3.C, contribution of each node goes from blue (weaker) to red (stronger)) in order to hold said mode.

For this study, Adjacency Matrices from over 88 were obtained with the assist of our tutor Rubén Calvo Ibañez.

In Figure 3.A, B, C we showcase said information for a single patient, which gives an intuition on the structure of study.

In Figure 3.B the two hemispheres of the brain (top left, bottom right) are depicted, showing strong intra-hemispheric correlations. Weak inter-hemispheric interactions are also present and are represented by the off-diagonal blocks (bottom-left and top-right).

As said previously, in Figure 3.C we obtain some of the connectome modes for a single patient. Extending this analysis to all patients, we find that the usual architecture of the brain, i.e., healthy patients, will show the same distribution of eigenvalues (from matrix $\mathbf{A}$) and the same distribution of degrees (from matrix $\mathbf{D}$). In this sense, this universality of the distribution may act as an indicator for healthy individuals, where a pathology can deviate the eigenvalues from the usual healthy ones.

However, it is important to note that this observation is not conclusive: different, unrelated systems can exhibit similar spectral distributions, and thus these features alone are insufficient to uniquely characterize brain health without additional structural/functional constraints. Nonetheless, this kind of analysis provides a starting point for identifying general principles behind brain network organization and its deviations for patho-

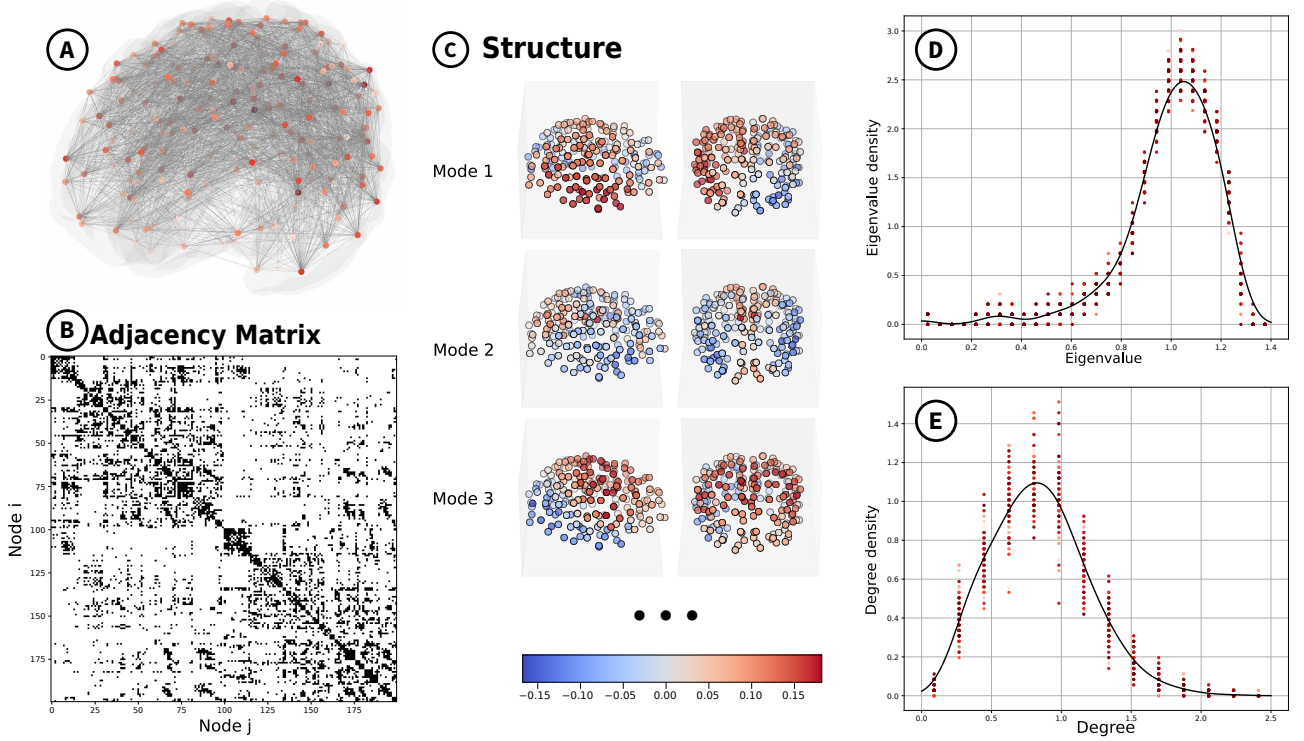


FIG. 3: **Structure of the brain:** (A) Illustration of the nodes studied within the brain, with the intensity indicating the number of connections to said node. (B) Representation of the Adjacency Matrix, the axis represent the nodes. A black dot between nodes represents a connection (1), a blank dot, no connection (0). In the plot, we grasp four distinct regions. The top left and the bottom right represent the two hemisphere on the brains, which are highly self-correlated. The top right and bottom left are the connections between said hemisphere that, although less connected, still holds interactions. (C) Connectome modes of the brain. (D) Eigenvalue histogram of the diagonalized connectome Laplacian for over 88 patients. (E) Degree histogram of the Adjacency matrix for over 88 patients. Both (D) and (E) exhibit universal distributions across healthy patients, indicating that these patterns remain consistent among individuals with structurally usual brains.

logical states.

C. Dynamics of the brain: correlations and the Kuramoto parameter

In this section, we try to obtain the similar dynamical regions like the ones seen in IIA for the structure obtained in IIB. In order to quantify how the different zones of the brain are related, we make use of the average Kuramoto order parameter r , defined as:

$$r e^{i\psi} = \frac{1}{N} \sum_{j=1}^N e^{i\theta_j}, \quad (11)$$

where θ_j indicates the phase for the j -th oscillator. Thus r tells us how correlated the nodes are in phase, which is just the degree of synchronization.

Studying this parameter as a function of the excitatory synaptic weights reveals the different dynamical regimes that arise in the full scope of the brain. Similar to the

single region obtained in Sec. IIA, see FIG. 4. Interestingly enough, the dynamical phase space obtained is similar but not equal to the one we analyzed for a single region, i.e., the coupling of different regions of the brain is not equal to the sum of the individual regions, *more is different*.

In order to better understand the fixed points featured on the right panel of Figure 4, we can create a parallel with the study done for a single region. From Eq.(6) and Eq.(7), the sum over the different nodes can be interpreted as the external inputs $P_{E,i}(t)$ and $P_{I,i}(t)$ dependent on time. Then, the fixed points for each of the regions will change as long as the rest of nodes are evolving in time. This is why we obtain a different dynamical landscape, as we have the original single equations plus a time-dependent external input (see eq.12 and eq.13), which is just a consequence of having interactions with other nodes.

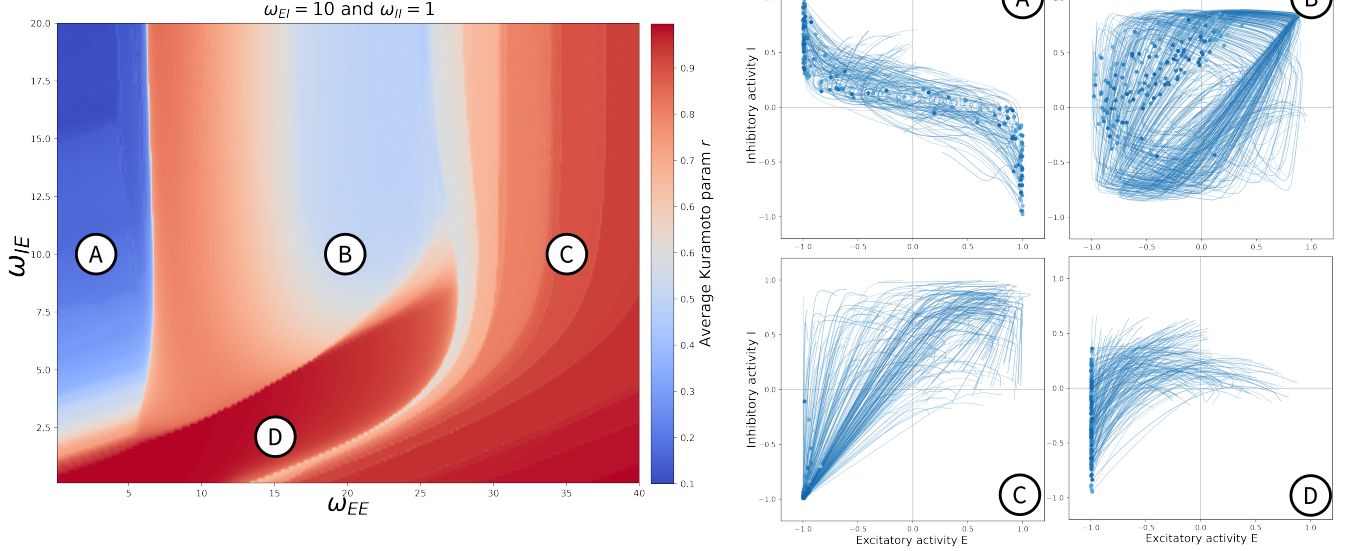


FIG. 4: **Dynamics of the brain:** Left: Average Kuramoto parameter r in the excitatory synaptic weights phase diagram, which shows four distinct regimes. Right: Trajectories in the phase space of the average firing rates E, I , each dot represents a node: **(A)** Quiescent: each node will tend to the fixed point of its region, **(B)** Oscillatory: several limit cycles are obtained, **(C)** & **(D)** Bistable, with different number of fixed points.

$$\tau_E \frac{dE_i}{dt} = -E_i + S(w_{EE}E_i - w_{EI}I_i + P_{E,i}), \quad (12)$$

$$\tau_I \frac{dI_i}{dt} = -I_i + S(w_{IE}E_i - w_{II}I_i + P_{I,i}), \quad (13)$$

with,

$$P_{E,i} = \sum_{j \neq i} (A_{ij}w_{EE}E_j - w_{EI}I_j) + P_E, \quad (14)$$

$$P_{I,i} = \sum_{j \neq i} (A_{ij}w_{IE}E_j - w_{II}I_j) + P_I. \quad (15)$$

Consequently, the dynamical regimes seen in Figure 4.A, C and D, will settle into equilibrium fixed points over time, which will correspond to the fixed points of the single regions of each node. As for the oscillatory regime, see Figure 4.B, we obtain several limit cycles.

III. CONCLUSIONS

In this study, the Wilson-Cowan model has been extensively studied, going from a single region model to a multiregional model corresponding to different regions of the brain.

For the single model, three dynamical regimes and four bifurcations between them have been found and studied.

In the study of structure, we found that decomposing the Laplacian of the human connectome reveals a surprisingly simple and almost universal structure that helps explain how functional networks in the brain are organized.

Because the connectome harmonics are orthogonal (and therefore independent) they form a complete set of building blocks. This makes them a powerful way to represent and study any pattern of brain activity across space and time.

Moreover, we have found that, between the healthy patients, that is, the ones with a typical brain architecture, there exists an ubiquitous distribution of the eigenvalues of the Adjacency matrix. This finding may be useful in order to detect a specific pathology that alters the structure of the brain, which could be found by analyzing if the corresponding eigenvalues are far away from the usual distribution seen in other patients. However, this is just a starting point, not a conclusive statement, for such task, more constraints and a more detailed analysis would be needed.

Studying the dynamical regimes for the full scope of the brain, we find a similar but not equal phase diagram as the one found for the single region, due to interregional connections. Therefore, we have obtained a complete framework for the study of the brain, gauged by the Kuramoto parameter, which gives an intuition of the synchronization between the different brain regions and how, depending on the degree of correlation, different global regimes arise.

Additionally, we have studied the Wilson-Cowan stochastic dynamics (see B), where we have obtained the expected stationary distribution around the mean field E .

In conclusion, in this study we have analyzed a powerful tool for the characterization of the brain's structure and dynamics, applying it to real data. Future work

would involve characterizing the exact functional form between dynamics and structure.

ACKNOWLEDGMENTS

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CODE AVAILABILITY

All the code used for carrying out the simulations can be found in our Github page.

Appendix A: Analytical bifurcations

In order to find the bifurcations, we will set the system's equations to zero, finding two implicit fixed points:

$$E^* = \tanh(w_{EE}E^* - w_{EI}I^*) = \tanh(h_E) \quad (\text{A1})$$

$$I^* = \tanh(w_{IE}E^* - w_{II}I^*) = \tanh(h_I) \quad (\text{A2})$$

Hopf Bifurcation:

The conditions for a Hopf bifurcation are:

$$\text{tr}(J) = 0, \quad \det(J) > 0 \quad (\text{A3})$$

The Jacobian is:

$$J = \begin{pmatrix} -1 + w_{EE}\gamma & -w_{EI}\gamma \\ w_{IE}\delta & -1 - w_{II}\delta \end{pmatrix} \quad (\text{A4})$$

With:

$$\gamma = 1 - (E^*)^2, \quad \delta = 1 - (I^*)^2 \quad (\text{A5})$$

Forcing the nullity of the trace:

$$\text{tr}(J) = 0 \rightarrow w_{EE} = \frac{2 + w_{II}(1 - (I^*)^2)}{1 - (E^*)^2} \quad (\text{A6})$$

In the trivial fixed point $E^* = I^* = 0$ and $w_{II} = 1$, we obtain the constant Hopf bifurcation:

$$w_{EE} = 3 \quad (\text{A7})$$

Pitchfork Bifurcation:

The condition for a Pitchfork bifurcation is:

$$\det J = 0 \quad (\text{A8})$$

And once again, considering the trivial fixed point $(0, 0)$ we arrive to:

$$w_{IE} = \frac{(w_{EE} - 1)(1 + w_{II})}{w_{EI}} = \frac{w_{EE} - 1}{5} \quad (\text{A9})$$

The other bifurcations are not obtainable analytically.

Appendix B: Stochastic brain dynamics

Let's recover the Wilson–Cowan equations from a microscopic point of view. The master equation for a population of n excitatory neurons and n inhibitory neurons is:

$$\begin{aligned} \frac{dP(t)}{dt} = & \alpha_E(n_E + 1)P_{n_E+1, n_I} \\ & - \alpha_E n_E P_{n_E, n_I} \\ & + (N_E - n_E + 1)f_E[s_E(n_E - 1, n_I)]P_{n_E-1, n_I} \\ & - (N_E - n_E)f_E[s_E(n_E, n_I)]P_{n_E, n_I} \\ & + \alpha_I(n_I + 1)P_{n_E, n_I+1} \\ & - \alpha_I n_I P_{n_E, n_I} \\ & + (N_I - n_I + 1)f_I[s_I(n_E, n_I - 1)]P_{n_E, n_I-1} \\ & - (N_I - n_I)f_I[s_I(n_E, n_I)]P_{n_E, n_I} \end{aligned} \quad (\text{B1})$$

In order to apply the Kramers–Moyal expansion, we need to take this discrete master equation to the continuum. For simplicity, let's take first the excitatory term and do the full derivation. For the inhibitory, it will yield the same results.

$$\begin{aligned} \frac{dP_n(t)}{dt} = & \alpha [(n + 1)P_{n+1}(t) - nP_n(t)] \\ & + (N - n + 1)f[s_E(n - 1)]P_{n-1}(t) \end{aligned} \quad (\text{B2})$$

Let's take it to the continuum by dividing everything by N and taking:

- $x = \frac{n}{N}$
- $P_n(t) \rightarrow P(x, t)$
- $n \rightarrow n \pm 1$ implies $x \rightarrow x \pm \Delta x$
- First term: $(n + 1)P_{n+1}(t) - nP_n(t) = N(x + \Delta x)P(x + \Delta x, t) - NxP(x, t)$
- Second term: $(N - n + 1)f[s_E(n - 1)]P_{n-1}(t) = N(1 - x + \Delta x)f(x - \Delta x)P(x - \Delta x)$

- Third term: $(N - n)f[s_E(n)]P_n(t) = N(1 - x)f(x)P(x, t)$

Now, we do a Taylor expansion in terms of Δx .

First term:

$$(x + \Delta x)P(x + \Delta x, t) - NxP(x, t) \approx xP(x) + \Delta x \partial_x [xP(x)] + \frac{\Delta x^2}{2} \partial_x^2 [xP(x)] - xP(x) + \mathcal{O}(\Delta x^3) \quad (\text{B1})$$

Second term:

$$(1 - x + \Delta x)f(x - \Delta x)P(x - \Delta x) \approx (1 - x)f(x)P(x) - \Delta x \partial_x [(1 - x)f(x)P(x)] + \frac{\Delta x^2}{2} \partial_x^2 [(1 - x)f(x)P(x)] \quad (\text{B2})$$

Now, summing all three contributions and canceling the zeroth-order terms, we get the effective evolution equation:

$$\partial_t P(x, t) = -\Delta x \partial_x [-\alpha x + (1 - x)f(x)] P(x, t) + \frac{\Delta x^2}{2} \partial_x^2 [\alpha x + (1 - x)f(x)] P(x, t) + \mathcal{O}(\Delta x^3) \quad (\text{B3})$$

Taking the continuum limit $\Delta x = \frac{1}{N} \rightarrow 0$, we obtain the Fokker-Planck equation:

$$\partial_t P(x, t) = -\partial_x [A(x)P(x, t)] + \frac{1}{2N} \partial_x^2 [B(x)P(x, t)] \quad (\text{B4})$$

with

$$A(x) = -\alpha x + (1 - x)f(x), \quad B(x) = \alpha x + (1 - x)f(x)$$

Comparing with the standard Fokker-Planck form:

$$\partial_t P(x, t) = -\partial_x [q(x)P(x, t)] + \frac{1}{2} \partial_x^2 [g(x) \partial_x [g(x)P(x, t)]]$$

we identify:

$$q(x) = A(x), \quad g^2(x) = \frac{B(x)}{N}$$

And the first moment, $A(x)$, represents the mean change per unit of time in x . Let's remember that x , in the mean field approximation, is $x = \frac{\langle n \rangle}{N}$, which is just the excitatory mean firing rate E . We recover the Wilson-Cowan equation through $A(x) \equiv \frac{dE}{dt}$:

$$A(x) = \frac{dE}{dt} = -\alpha E + (1 - E)f(E) \quad (\text{B5})$$

Where $f(x)$ is the sigmoid function and the $(1 - E)$ is an effect of considering finite population. For the inhibitory population, the treatment is completely analogous and coupling both equations will give the original phenomenological Wilson-Cowan model.

To derive the corresponding stochastic differential equation, we introduce the q and g^2 terms in the associated Itô stochastic differential equation:

$$\dot{x}(t) = A(x)t + \sqrt{\frac{B(x)}{N}} \xi(x, t), \quad (\text{B6})$$

where $\xi(x, t)$ is Gaussian white noise, representing noise from finite population effects.

The deterministic part of Eq. (B6), $\dot{x} = A(x)$, reproduces the mean-field Wilson-Cowan dynamics for the excitatory population.

The stochastic term $\sqrt{B(x)/N} \xi(x, t)$ introduces fluctuations around the deterministic trajectory, whose magnitude scales inversely with the system size N . The trajectories over time can be seen in Figure 5.

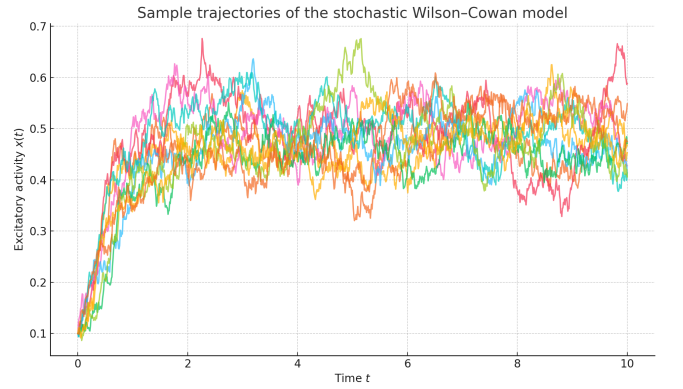


FIG. 5: Sample trajectories for the excitatory stochastic Wilson-Cowan model. With the initial condition of $x = 0.1$, the system will tend to the mean field expected value, averaged with noise due to the stochastic component.

A stationary distribution is acquired for enough samples, in Figure 6.

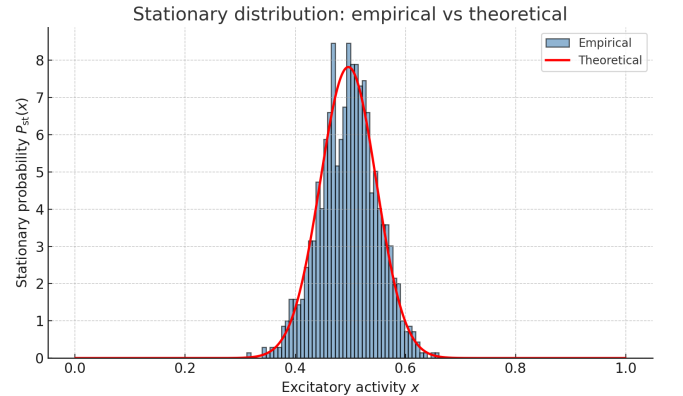


FIG. 6: Theoretical and empirical stationary distribution of the excitatory stochastic Wilson-Cowan model.

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